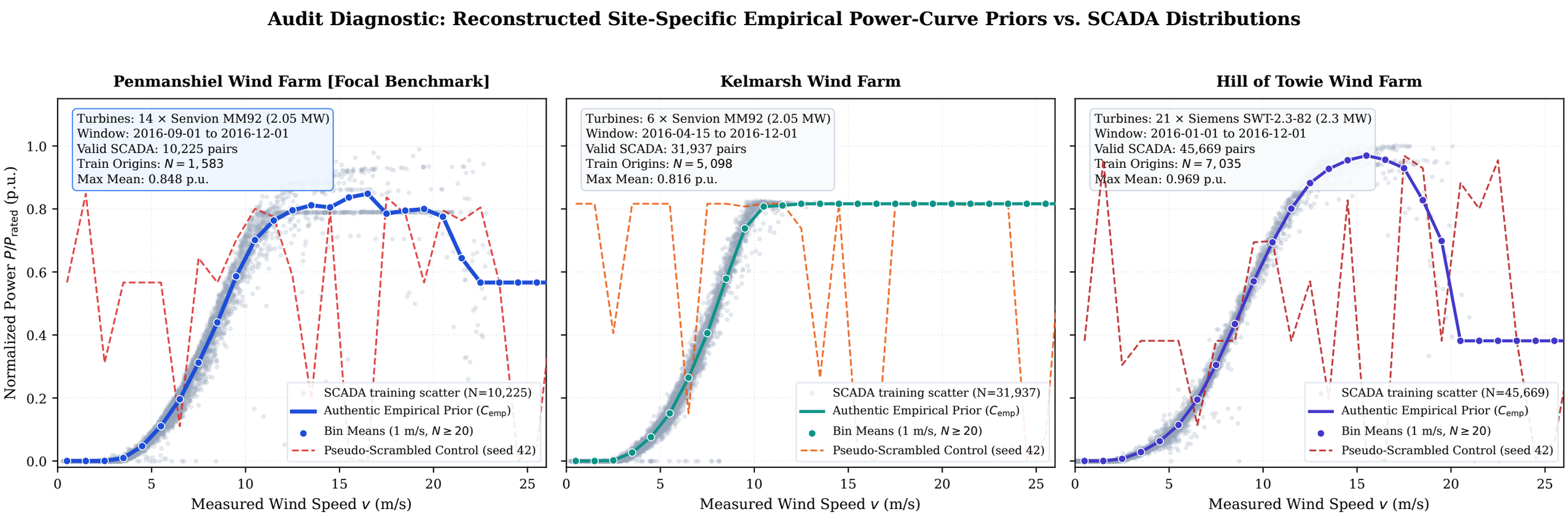

Audit Diagnostic: Reconstructed Site-Specific Empirical Power-Curve Priors vs. SCADA Distributions
Penmanshiel Wind Farm [Focal Benchmark]
Kelmarsh Wind Farm
Hill of Towie Wind Farm
Turbines: 14 × Senvion MM92 (2.05 MW)
Window: 2016-09-01 to 2016-12-01
Valid SCADA: 10,225 pairs
Train Origins: N = 1,583
Max Mean: 0.848 p.u.
Turbines: 6 × Senvion MM92 (2.05 MW)
Window: 2016-04-15 to 2016-12-01
Valid SCADA: 31,937 pairs
Train Origins: N = 5,098
Max Mean: 0.816 p.u.
Turbines: 21 × Siemens SWT-2.3-82 (2.3 MW)
Window: 2016-01-01 to 2016-12-01
Valid SCADA: 45,669 pairs
Train Origins: N = 7,035
Max Mean: 0.969 p.u.
SCADA training scatter (N=10,225)
SCADA training scatter (N=31,937)
SCADA training scatter (N=45,669)
Authentic Empirical Prior (C_emp)
Bin Means (1 m/s, N ≥ 20)
Pseudo-Scrambled Control (seed 42)
Normalized Power P/P_rated (p.u.)
Measured Wind Speed v (m/s)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25